SILVER OAK UNIVERSITY
TUTORIAL – 10
Automation & Regression Analysis

Aim: Analyze black-box testing, white-box testing, integration testing, and system testing on a sample web application.

1. Implementation Details

Parameter	Details
Project Name	E-Commerce Web Application Testing
Application Software	SauceDemo (Sample E-Commerce Web Application)
Module Name	Login, Product Catalog, Cart, Checkout, Order Confirmation
SDLC Model	Agile
Programming Language	Java
IDE Tool Used	IntelliJ IDEA
Input/Test Data	Valid/invalid login data, product selection, cart quantity, checkout details

2. Module Analysis Table

Module Name	Functionality	Critical (Yes/No)	Frequently Changed (Yes/No)	Suitable for Automation (Yes/No)	Reason
Login	User authentication and logout	Yes	Yes	Yes	Critical and repetitive; ideal for regression automation.
Product Catalog	View products, sort and select product	Yes	Yes	Yes	Stable, repetitive UI actions and frequent regression checks.
Shopping Cart	Add/remove products and verify cart	Yes	Yes	Yes	Business-critical and repeatedly executed.
Checkout	Enter customer details and continue	Yes	Yes	Yes	Critical purchase flow; suitable for automated regression.
Order Confirmation	Complete order and verify confirmation	Yes	No	Yes	Important end-to-end validation with repeatable expected results.

3. Automation Candidate Identification

Test Case ID	Functionality	Automation Recommended (Yes/No)	Reason	Automation Tool
TC-01	Valid Login	Yes	Repeated on every regression cycle	Selenium WebDriver
TC-02	Invalid Login Validation	Yes	Stable negative test and easy to repeat	Selenium WebDriver
TC-03	Add Product to Cart	Yes	Frequent and business-critical	Selenium WebDriver
TC-04	Checkout Flow	Yes	Long repetitive workflow; saves execution time	Selenium WebDriver + TestNG
TC-05	Order Confirmation	Yes	Repeatable end-to-end regression scenario	Selenium WebDriver + TestNG

4. Test Case Detailed

Detailed test case documentation is provided for the representative critical test case TC-01. Detailed execution tables do not need to be repeated for every test case; the remaining cases are documented in the Test Case Summary and Regression Test Suite.

Field	Details
Test Case ID	TC-01
Requirement ID	REQ-01
Module Name	Login
Feature Under Test	Valid User Authentication
Test Scenario	Login with valid credentials

Test Objective	Verify that a registered user can log in successfully.
Test Type (Unit/Integration/System/UAT)	Integration
Test Technique (Black Box/White Box)	Black Box
Priority	High
Severity	Critical
Preconditions	Valid user account exists; application is available.
Postconditions	Authenticated user is on the inventory page.
Test Environment	Web application testing environment
Browser/OS (if applicable)	Google Chrome / Windows 11
Test Data	Username: standard_user; valid password
Expected Result	Inventory page is displayed after successful authentication.
Actual Result	Inventory page displayed successfully.
Status (Pass/Fail)	Pass
Executed By	Student
Execution Date	07-09-2026

5. Test Case Execution

Step No.	Test Steps	Input	Expected Result	Actual Result	Status
1	Open the application login page	SauceDemo URL	Login page is displayed	Login page displayed	Pass
2	Enter valid username	standard_user	Username is accepted	Username accepted	Pass
3	Enter valid password	Valid password	Password is accepted	Password accepted	Pass
4	Click Login	Login button	User is authenticated and inventory page opens	Inventory page opened	Pass
5	Verify inventory page title/products	Products page	Inventory page and product list are visible	Product list visible	Pass

6. Regression Test Suite

Regression Test ID	Module	Scenario	Expected Result	Status
RT-01	Login	Valid login	User reaches inventory page	Pass
RT-02	Login	Invalid login	Error message displayed	Pass
RT-03	Product Catalog	Add product to cart	Selected product appears in cart	Pass
RT-04	Shopping Cart	Remove product	Product removed from cart	Pass
RT-05	Checkout	Submit valid checkout information	Checkout overview displayed	Pass
RT-06	Order	Complete order	Order confirmation displayed	Pass

7. Automation Feasibility Report

Criteria	Observation	Value	Recommendation
Repetitive Test Cases	TC-01 to TC-06 are repeatedly executed	High	Automate
Stable Functionality	Login, cart and checkout workflows are stable	High	Automate
High Business Priority	Login and checkout directly affect purchasing	High	Automate
Execution Time	Automated regression reduces repeated manual effort	High	Automate
Automation Tool	Selenium WebDriver with Java/TestNG is suitable	High	Recommended

8. Regression Testing Summary

Metric	Value
Total Modules Analyzed	5
Automation Candidates	5
Regression Test Cases	6
Modules Excluded	0
Recommendation	Automate stable, repetitive, high-priority regression scenarios.

9. Observation & Remarks

Sr. No.	Observation	Remarks
1	Login and authentication were validated with positive and negative scenarios.	Login flow is suitable for regression automation.
2	Product selection and cart operations were checked.	These flows are repetitive and business-critical.
3	Checkout information was validated using valid test data.	Checkout should be part of every regression cycle.
4	Order completion was checked through the confirmation page.	End-to-end automation is recommended.
5	Regression suite covered major customer purchase paths.	Selenium WebDriver + TestNG is recommended for repeatable execution.

Conclusion

The sample e-commerce application was analyzed using black-box, integration and system-level testing concepts. The login, product, cart, checkout and order modules are suitable for automation because they are repetitive, business-critical and have predictable expected results. Selenium WebDriver with Java and TestNG is recommended for the regression suite. Only one representative test case (TC-01) is expanded into the detailed test-case and step-execution format, while the other cases are covered in the test-case summary and regression suite.

Note: The Pass/Fail entries above are sample lab execution data prepared for the report. Replace them with your actual observations/screenshots if your instructor requires evidence from a live run.